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BATTERY MODULE FOR WORK MACHINE

Abstract

A battery module is disclosed. The battery module comprises: a frame; a plurality of lithium-ion cells provided in the frame; a compression plate attached to the frame and configured to apply a uniform compression pressure to the plurality of lithium-ion cells in the frame; and a stopper for holding the compression plate at a holding location for maintaining pressure.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to battery modules, and more particularly relates to compression of lithium ion cells in battery modules.

BACKGROUND

[0002] Lithium-ion (Li-ion) cells, integral to numerous modern technologies, are available in various forms, including cylindrical, prismatic, and pouch cells. Each form exhibits distinct structural and operational characteristics. Cylindrical cells are particularly noted for their robustness against cyclic and through-life expansions. In contrast, prismatic, blade, pouch, or similar cells, due to their construction, do not exhibit the same resilience, posing challenges in terms of durability and longevity. The assembly and operational efficacy of prismatic, blade, pouch, or similar cells Li-ion cells are heavily contingent on maintaining optimal compression. This requirement is critical to ensure safety, reliability, and maximum life of these cells.

[0003] Currently, this is achieved through a cell stack structure predominantly comprising two end plates and various restraining members. These restraining members are responsible for maintaining the necessary compression between the endplates and come in different forms, such as sheets welded or riveted to endplates, straps wound around endplates, metal strips strapped over endplates, and tie rods used to bolt two endplates together.

[0004] Despite the critical role of compression in the performance of prismatic, blade, pouch, or similar cells, the current methods of achieving this compression exhibit several shortcomings. The “box in box” construction of the battery module, a common approach in current designs, leads to a reduction in both volumetric and gravimetric energy density. This construction style also results in increased thermal resistance between the cell surface and cooling plate or, alternatively, a reduction of the cooling surface due to the restraining members. Additionally, this approach increases the complexity of the design, evidenced by an increased number of parts and a lack of precise control over the compression force. This lack of control is often due to the variability introduced by factors such as the restraining member's settling stress under tension.

[0005] Existing designs of battery modules in work machines suffer from inefficiencies due to their complex construction leading to decreased energy densities and challenges in thermal management, exacerbated by the increased number of components and the inherent design's inability to precisely control compression forces. The inability to control compression forces in the battery module reduces performance and reliability in existing battery modules used in work machines due to demanding operational environments of heavy-duty work machinery like excavators.

[0006] Others have attempted to develop an effective design for Li-ion compression, but fail to provide an efficient design for maintaining Li-ion compression. For example, U.S. Pat. No. 9,559,378 discloses a fuel cell that includes unit cells that are stacked, a case that houses the cell stack, and a pressure plate that is placed in the case in the stacking direction to compress the cell stack by a plurality of load adjusting screws. The use of multiple load adjusting screws in U.S. Pat. No. 9,559,378 for compressing the cell stack introduces complexity and inefficiency. Manual tuning of each screw is time-intensive and may result in uneven pressure, hindering uniform compression. This approach struggles to uniformly address cell expansion and contraction, potentially causing inconsistent stress distribution and impacting the cells' longevity and performance.

[0007] It can therefore be seen that a need exists for an improved solution in Li-ion cell compression design that simplifies the compression process, ensures uniform pressure distribution, and accommodates the natural expansion and contraction of cells during their lifecycle.

SUMMARY

[0008] In accordance with one aspect of the disclosure, a battery module is disclosed. The battery module comprises: a frame; a plurality of lithium-ion cells provided in the frame; a compression plate attached to the frame and configured to apply a uniform compression pressure to the plurality of lithium-ion cells in the frame; and a stopper for holding the compression plate at a holding location for maintaining pressure.

[0009] In accordance with one aspect of the disclosure, a system is disclosed. The system comprises a work machine; and a battery module for powering the work machine. The battery module includes: a frame; a plurality of lithium-ion cells provided in the frame; a compression plate attached to the frame and configured to apply a uniform compression pressure to the plurality of lithium-ion cells in the frame; and a stopper for holding the compression plate at a holding location for maintaining pressure

[0010] In accordance with another aspect of the disclosure, a method for assembling a lithium-ion battery module is disclosed. The method comprises positioning a plurality of lithium-ion cells within a frame of a battery module; installing a front cover on a first end of the frame, a back cover on a second end of the frame, and a compression plate between the plurality of lithium-ion cells and the back cover; and securing the compression plate in place, via a stopper, to apply a compression pressure on the plurality of lithium-ion cells to mitigate bulging of the plurality of lithium-ion cells within the frame.

[0011] These and other aspects and features of the present disclosure will be better understood upon reading the following detailed description when read in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. **1** is a perspective view of a work machine, according to an embodiment of the present disclosure.

[0013] FIG. **2** is a perspective view of a battery module with a single column, according to an embodiment of the present disclosure.

[0014] FIG. **3** is a perspective cut-away view of a Li-ion cell for use in the battery module of FIG. **2**, according to an embodiment of the present disclosure.

[0015] FIG. **4** is a pack of Li-ion cells of FIG. **3** for use in the battery module of FIG. **2**, according to an embodiment of the present disclosure.

[0016] FIG. **5** is a compression plate for use in the battery module of FIG. **2**, according to an embodiment of the present disclosure.

[0017] FIG. **6** is a perspective cut-away view of the battery module taken along line **6-6** of FIG. **2** containing a pack of Li-ion cells, according to an embodiment of the present disclosure.

[0018] FIG. **7** is a cross-sectional view of the of the battery module of FIG. **2**, taken along line **7-7** of FIG. **6**, according to an embodiment of the present disclosure.

[0019] FIG. **8** is a shim for use in the battery module of FIG. **2**, according to an embodiment of the present disclosure.

[0020] FIG. **9** is a perspective view of the battery module of FIG. **2**, according to another embodiment of the present disclosure.

[0021] FIG. **10** is a perspective close up of an end of the battery module of FIG. **2**, according to an embodiment of the present disclosure.

[0022] FIG. **11** is a top cross-sectional view of a multi-pack battery module, according to an embodiment of the present disclosure.

[0023] FIG. **12** is a perspective cut-away view of the multi-pack battery module taken along line **12-12** of FIG. **11**, according to another embodiment of the present disclosure.

[0024] FIG. **13** is a flow-chart of a method of assembling a lithium-ion battery module, according to an embodiment of the present disclosure.

[0025] FIGS. **14** is a perspective view of the frame of the battery module of FIG. **1**, according to another embodiment of the present disclosure.

[0026] FIGS. **15** is a perspective view of the frame of the battery module of FIG. **11**, according to another embodiment of the present disclosure.

[0027] The figures depict one embodiment of the presented disclosure for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

DETAILED DESCRIPTION

[0028] Referring now to the drawings, and with specific reference to the depicted example, a work machine **10** is shown, illustrated as an exemplary excavator. Excavators are heavy equipment designed to move earth material from the ground or landscape at a dig site in the construction and agricultural industries.

[0029] Referring now to FIG. **1**, the work machine **10** comprises a frame **12**. The frame **12** is supported on ground engaging elements **14**, illustrated as continuous tracks. It should be contemplated that the ground engaging elements **14** may be any other type of ground engaging elements **14** such as, for example, wheels, etc. The work machine **10** further includes a prime mover **16** in the frame **12**, a work implement **18** extending from the frame **12** for conducting work with a work tool **20**, such as, for example, excavating landscapes or otherwise moving earth, soil, or other material at a dig site, and a cab **22** for operator personnel to operate the work machine **10**. The prime mover **16** may be an engine or electric motor, as generally known in the arts.

[0030] The work machine **10** may further comprise a battery module **100**, exemplified as a Lithium-ion battery assembly, for powering operations of the work machine **10**, the ground engaging elements **14**, the prime mover **16**, and the work implement **18**. The battery module **100** may replace the prime mover **16** or be provided in conjunction with the prime mover **16** in the work machine **10**.

[0031] Lithium-ion battery assemblies are critical in providing power for a wide range of applications, from portable electronics to electric vehicles and renewable energy systems. While the following detailed description illustrates an exemplary embodiment in connection with a standard energy storage system, it should be understood that the principles and applications of the present disclosure extend to various other systems, devices, including, but not limited to, portable electronic devices, electric vehicles, industrial machinery, and grid storage solutions, among others. Furthermore, while the following detailed description describes an exemplary aspect of the battery module **100** in connection with the excavator, it should be appreciated that the description applies equally to the use of the present disclosure in other work machines, including, but not limited to, backhoes, front-end loaders, shovels, draglines, skid steers, wheel loaders, and tractors, as well.

[0032] Referring now to FIG. **2**, the battery module **100**, is shown in a perspective view, according to one embodiment of the disclosure. The battery module **100** comprises a frame **102** that serves as the foundational framework for the battery module **100**. The frame **102** may include a single column that is designed to house multiple types of Li-ion cells and to facilitate the compression and maintenance of said cells within a single column formed by the frame **102**. The frame **102** may be formed in a module structure and as a unitary frame to help reduce leakage by having a uniform frame.

[0033] The frame **102** is engineered in a single column to accommodate a variety of lithium-ion cells, addressing the need for versatility in application. Additionally, the battery module **100** includes a front cover **104** and a top cover **106**, offering protection and maintaining the integrity of the internal components. The battery module **100** also comprises battery terminals **108**, including both positive and negative terminals, essential for establishing electrical connections, as generally known in the arts. The design of the battery module **100** facilitates ease of assembly and enhanced compression capabilities. This is particularly significant in addressing the common issue of bulging **306** in lithium-ion batteries for improved performance and energy longevity in the battery module **100**. The structure of the battery module **100** structure ensures that the Li-ion cells **300** are held securely, mitigating the risks associated with cell expansion over time. The frame **102** may be made from steel, metal, or other material that maintains the structure of the frame **102** to limit the bulging

306, as generally known in the arts. The front cover **104** and top cover **106** may be welded, bolted, soldered, or otherwise sealed to the frame **102**, as generally known in the arts.

[0034] Now referring to FIG. **3**, a perspective cut-away of a Li-ion cell **200**, according to an embodiment of the disclosure. The Li-ion cell **200** includes a sub-ion-cell **202** including an anode **204** and a cathode **206**, separated by a spacer **208**. An ion sheet **210** is included to facilitate the movement of ions within the Li-ion cell **200**. The Li-ion cell **200** is encased in a shell **212**, which ensures physical protection and chemical integrity. A terminal cover **214** safeguards the electrical terminals of the Li-ion cell **200**, crucial for maintaining a stable and secure connection within the battery module **100**.

[0035] The Li-ion cell **200** can be configured in different forms such as prismatic and pouch cells. Prismatic cells are known for their rigid, rectangular casing that offers robustness and efficient space utilization, making them a favorable choice for applications where space and durability are concerns. Pouch cells, on the other hand, are characterized by their flexible, foil-like casing. This design allows for a lighter weight and more flexible form factor in the shell **212**, making them suitable for applications where weight and design flexibility are crucial. Both prismatic and pouch cells offer unique advantages and can be selected based on the specific requirements of the application of the battery module **100** or availability in the market.

[0036] Now referring to FIG. **4**, a pack of Li-ion cells **300** is illustrated, according to an embodiment of the disclosure. This pack of Li-ion cells **300** comprises a plurality of the Li-ion cell **200**, each featuring a positive cell terminal **302** and a negative cell terminal **304**. A variety of specifications may be set forth by different manufacturers that manufacture the Li-ion cell **200**. These specification may include initial compression requirements and end-of-life compression requirements for optimal battery performance. The specifications may differ from varying manufacturers the Li-ion cell **200** and the pack of Li-ion cells **300**.

[0037] The Li-ion cell **200** and the pack of Li-ion cells **300** often face the challenge of bulging **306**, typically caused by the buildup of gases inside the Li-ion cell **200** during charging and discharging cycles. This bulging **306** is a significant concern as it can lead to structural deformation of the Li-ion cell **200**, potential breaches of the Li-ion cell **200** and the pack of Li-ion cells **300**'s integrity, and in extreme cases, safety hazards to the battery module **100**.

[0038] Now referring to FIG. **5**, a compression plate **400** within the battery module **100** is illustrated, according to an embodiment of the disclosure. The compression plate **400**, in conjunction with a stopper **402**, is positioned in the battery module **100** to apply a uniform force on the pack of Li-ion cells **300** within the module. This uniform application of force is essential in managing the expansion tendencies of these cells, commonly observed during their operational lifecycle. The design of the compression plate **400** is tailored to align with the specific dimensions and configurations of the battery module **100**, ensuring effective integration and functionality. This alignment aids in maintaining the stability and operational efficiency of the li-ion cells **300**, thereby enhancing the overall performance and extending the service life of the cells. The stopper **402** may extend from the compression plate **400**, as a cylinder, rectangular, or other shape generally known in the arts. Pressure may be applied to the stopper **402** to place the compression plate **400** in a desired position. Pressure may be applied to a free end of the stopper **402** to compress the pack of Li-ion cells **300** via the compression plate **400**. The free end may be sealed to the battery module **100** when the compression plate **400** is in a preferred position applying a minimum compression pressure for the beginning and end-of-life of the battery module **100**.

[0039] Now referring to FIG. **6**, a perspective cut-away view of the battery module **100** taken along line **6-6** of FIG. **2** is illustrated, according to an embodiment of the disclosure. FIG. **6** illustrates the internal structure of the battery module **100**, as well as the configuration and placement of the plurality of the Li-ion cells **300** within the frame **102**. The assembly of these Li-ion cells **300** is strategically designed to optimize both their individual and collective performance within the battery module **100**. Each cell in the pack of Li-ion cells **300** are aligned in a manner that can receive uniform distribution of pressure. The compression plate **400** is engineered to apply consistent and constant pressure across the pack of Li-ion cells **300**, which is essential for counteracting potential cell expansion that causes bulging **306** while ensuring consistent electrical output. This uniform pressure distribution is crucial for prolonging the operational life of the Li-ion cells **300** and maintaining the module's overall efficiency.

[0040] The frame **102**, the front cover **104**, and the top cover **106** are designed to encase the Li-ion cells **300** and related elements, providing a protective barrier against external environmental influences while maintaining compression for reducing the bulging **306**. The frame **102**, the front cover **104**, and the top cover **106** ensure that the internal components are shielded from physical impact and environmental conditions which could otherwise compromise their functionality and lifespan, such as air, dust, moisture, debris, etc.

[0041] The frame **102**, designed as a modular and unitary structure, enhances the overall integrity and efficiency of the battery module **100**. Constructed as a single, cohesive unit, the frame **102** minimizes potential points of weakness and leakage. The uniformity of the frame **102** is beneficial in maintaining a controlled environment for the packs of Li-ion cells **300** in a single column, ensuring that they are protected from external contaminants and physical damage that could compromise their performance.

[0042] When bulging **306** occurs, the unitary design of the frame **102** becomes advantageous by providing robust and seamless structure for a stable support system that can help distribute the stress caused by bulging **306** more evenly. This can mitigate the impact of bulging on the Li-ion cell **200** and the pack of Li-ion cells **300** in the battery module **100** as a whole, thereby enhancing the operational reliability and longevity of the battery assembly. Additionally, the modular nature of the frame **102** allows for adaptability and scalability, accommodating different sizes and configurations of lithium-ion cells as needed for various applications. This versatility, coupled with the strength of a unitary design, makes the frame **102** a critical component in advancing the durability and efficiency of lithium-ion battery modules.

[0043] Now referring to FIG. **7**, a cross-sectional view of the battery module **100** taken along line **7-7** of FIG. **6** is illustrated, according to an embodiment of the disclosure. A back cover **600** is provided to enclose the battery module **100**. The back cover **600**. The back cover **600** secures and protects the rear portion of the battery module **100** and holds the compression plate **400** in place for constant applied pressure against the pack of Li-ion cells **300**. This facilitates the distribution of a compression pressure **602** across the Li-ion cells **300**. The uniform application of the compression pressure **602** maintains the structural integrity of the plurality of the Li-ion cells **300** and preventing potential deformation or damage that could arise from uneven pressure distribution. Moreover, the back cover **600** serves to enclose and shield the internal components of the battery module **100** from external environmental factors, thus preserving the operational efficacy of the Li-ion cells **300**. The back cover **600** may be welded, sealed, soldered, bolted, or otherwise attached to the frame **102**, as generally known in the arts. A relief cap **604** is provided on the back cover **600** and configured to vent out gases from within the battery module **100** such as if the battery module **100** goes into a thermal runaway condition.

[0044] The stopper **402** may extend out through an aperture **606** in the back cover **600** to allow for a piston to apply pressure to meet the manufacture compression specifications of the pack of Li-ion cells **300** so that constant pressure is applied throughout the life of the battery module **100**. Once the required pressure is applied to the stopper **402**, the compression plate **400** can be maintained in place to provide a compression pressure **602** by bolting, welding, using circlip, or any mechanical method, as generally known in the arts. The back cover **600** includes the aperture **606** which is sealed by the stopper **402** serving as dust and water seal cover. The back cover **600** holds the compression plate **400** in place to provide the compression pressure **602**.

[0045] Now referring to FIG. **8**, a shim **700** used in the battery module **100** is

[0046] illustrated, according to an embodiment of the disclosure. The shim **700** may be a machined component and serves in supporting the compression pressure **602** against the pack of Li-ion cells **300** in the battery module **100**. The shim **700** is positioned between the compression plate **400** and the back cover **600**. The shim **700** assists in maintaining the position and effectiveness of the compression plate **400**. This placement ensures that the compression plate **400** remains securely in place, contributing to the consistent application of compression pressure **602** on the pack of Li-ion cells **300**.

[0047] Moreover, the shim **700** may be configured to adapt around the stopper **402** that extends from the compression plate **400**, allowing for a more tailored fit and efficient transmission of compressive force. By ensuring that the compression plate **400** is held firmly against the pack of Li-ion cells **300**, the shim **700** assists in the uniform distribution of pressure throughout the battery module **100** and supports managing the mechanical stresses imposed on the pack of Li-ion cells **300**. The integration of the shim **700** into the battery module **100** provides additional stability and operational reliability of the battery module **100**, especially in maintaining the desired compression levels over the lifespan of the pack of Li-ion cells **300** purchased from varying manufactures having varying compression requirements and specifications.

[0048] Now referring to FIG. **9**, a perspective view of the battery module **100** is illustrated with an electric busbar **800**, according to another embodiment of the disclosure. The electric busbar **800** is positioned atop the terminal cover **214** of each of the pack of Li-ion cells **300**, as generally known in the arts. The primary function of the electric busbar **800** is to facilitate the electrical connection between the positive and negative terminals of the Li-ion cells **300**. By linking the terminals across the pack of Li-ion cells **300**, the electric busbar **800** enables an efficient and organized electrical pathway throughout the battery module **100**. This arrangement is crucial for the coherent and effective operation of the Li-ion cells **300** within the battery module **100**, ensuring that electrical connectivity is maintained across the entire assembly.

[0049] Now referring to FIG. **10**, a perspective close-up of the back cover **600** is illustrated, according to an embodiment of the disclosure. The back cover **600** is designed to securely seal the rear of the battery module **100**, encapsulating the internal components, including the compression plate **400** and the stopper **402**. The stopper **402**, sealed to the back cover **600**, ensures that the compression plate **400** remains fixed in position by consistently applying compression to the Li-ion cells **300**. This consistent compression is essential for maintaining the structural integrity and optimal performance of the Li-ion cells **300**.

[0050] Now referring to FIG. **11**, a top schematic view of a multi-pack battery module **900** is illustrated, according to an embodiment of the disclosure. The frame **102** of the multi-pack battery module **900** is configured to hold a plurality of the pack of Li-ion cells **300** in a plurality of rows, optimizing space utilization and facilitating efficient energy storage. Each row containing a pack of Li-ion cells **300** is designed to fit snugly within the frame **102**, ensuring a compact and organized layout. This arrangement not only maximizes the capacity of the multi-pack battery module **900** but also contributes to the ease of assembly and maintenance.

[0051] For each pack of Li-ion cells **300**, a dedicated compression plate **400** may be provided in each row. This compression plate **400** is essential for applying uniform force across each pack of Li-ion cells **300**, thereby maintaining consistent pressure and minimizing the risk of bulging **306** or cell deformation. The compression plate **400**'s design is aligned with the specific dimensions of each row, ensuring that the pressure is distributed evenly and effectively across the Li-ion cells **300**.

[0052] Incorporated into each row, the shim **700** may be utilized to further enhance the stability and effectiveness of the compression pressure **602**. Positioned between the compression plate **400** and the Li-ion cells **300**, the shim **700** supports the maintenance of compression by holding the compression plate **400** in its desired position. This addition of the shim **700** is particularly beneficial in ensuring that the compression pressure is sustained over the lifecycle of the Li-ion cells **300**, catering to the varying compression specifications of different manufacturers of Li-ion products. The use of the shim **700** in each row thus plays a significant role in the overall functionality and longevity of the multi-pack battery module **900**, ensuring that each row of Li-ion cells **300** remains optimally compressed and functional.

[0053] Now referring to FIG. **12**, a perspective cut-away of the multi-pack battery module **900** taken along line **12-12** of FIG. **11** is illustrated, according to an embodiment of the disclosure. As shown in FIG. **12**, the frame **102** may be formed with multiple rows to accommodate a plurality of the pack of Li-ion cells **300**. The frame **102** may be formed with a plurality of cooling channels **902** to allow for coolant to be passed through the plurality of cooling channels **902** to cool the Li-ion cells **300** from the sides and bottom of the frame **102**. The plurality of cooling channels **902** may be incorporated into a separate cooling plate configured to the frame **102**.

Industrial Applicability

[0054] In practice, the innovative design detailed herein is applicable across a spectrum of industries, encompassing but not confined to sectors such as consumer electronics, electric vehicle manufacturing, energy storage, and portable power supply industries. Specifically, the module structures, Li-ion cell **200**, and compression methods of the present disclosure may be utilized in the development and enhancement of battery systems for devices ranging from handheld electronics to electric automobiles, as well as stationary energy storage systems. While the current disclosure addresses the application of Li-ion cells in battery assemblies, the principles and advancements outlined are transferrable to other battery technologies, including but not limited to nickel-metal hydride, solid-state batteries, and supercapacitors.

[0055] Now referring to FIG. **13**, a flowchart for a method **1000** of assembling the battery module **100** is illustrated, according to an embodiment of the disclosure. In a step **1002**, the method **1000** begins with positioning the pack of Li-ion cells **300** within the frame **102** of the battery module **100**. In step **1002**, the pack of Li-ion cells **300** are arranged within the designated compartments or row(s) in the frame **102** of the battery module **100** or the multi-pack battery module **900**, so that each Li-ion cell **200** fits securely and is aligned correctly for efficient electrical connectivity and uniform pressure application.

[0056] In a step **1004**, the method **1000** continues with the installation of a front cover **104** on a first end of the frame **102** and a back cover **600** on a second end of the frame **102**. Additionally, the compression plate **400** is installed between the pack of Li-ion cells **300** and the back cover **600**. The front cover **104** serves to protect the pack of Li-ion cells **300** from external environmental elements as well as providing battery terminals **108**, while the back cover **600** encloses the opposite end of the battery module **100**. The compression plate **600**'s placement provides the necessary compression pressure **602** against the pack of Li-ion cells **300**, ensuring their stability and mitigating bulging **306**.

[0057] In a step **1006**, the method **1000** secures the compression plate **400** to the back cover **600**, via the stopper **402** or by using the shim **700** so that the compression plate **400** remains in place to apply uniform pressure on the pack of Li-ion cells **300** for mitigating the bulging **306** of the Li-ion cell **200** within the frame **102**.

[0058] In the assembly of the lithium-ion cell, a piston may be utilized in precisely positioning the compression plate **400** against the pack of Li-ion cells **300**. This process begins by aligning the piston against the stopper **402**, which extends from the compression plate **400**. The piston is then activated to exert a controlled force on the stopper **402**, thereby moving the compression plate **400** into the desired position ensuring that the compression plate **400** applies the necessary uniform compression force to the Li-ion cells **300**. This force is calibrated to meet the specific compression requirements set forth by the manufacturer for optimal cell performance and longevity.

[0059] Once the compression plate **400** is correctly positioned, achieving the desired compression force, the stopper **402** is securely sealed by welding the stopper **402** to the back cover **600** or through the aperture **606** in the back cover **600**. This sealing process is vital to maintain the set position of the compression plate **400**, ensuring that the consistent pressure is applied to the Li-ion cells **300** throughout their operational life. Alternatively, the compression plate **400** may be held in position by mechanical fastening or by using the shim **700**. In some instances, based on the design and requirements of the battery module **100**, the extending part of the stopper **402** may be trimmed or cut post-sealing to maintain the compactness and integrity of the battery module **100** assembly. The use of a piston for precise positioning and sealing of the stopper **402** ensures that the battery module **100** maintains its structural integrity and functional efficacy, required for high-performance battery systems.

[0060] The modular approach to the frame **102**, such as U-shaped with a single row or with a plurality of rows **904**, allows for a high degree of customization to accommodate varying sizes and quantities of the Li-ion cell **200** and the pack of Li-ion cell **200**. The U-shaped configuration of the frame **102** is characterized by its open-ended structure, which is pivotal for easy insertion and removal of lithium-ion cells. The frame **102** simplifies the assembly process and also ensures that the cells are securely held in place, enhancing the structural integrity of the battery module **100**. Furthermore, the U-shaped frame **102** is designed to provide a support system for the Li-ion cells, with sides of the frame **102** extending upwards to form a protective enclosure and preventing lateral movement of the pack of Li-ion cells **300**, thereby minimizing the risk of physical damage and electrical shorts. The shape also enables the frame **102** to accommodate additional components, such as thermal insulators or cooling mechanisms, within its structure, along the interior sides, enhancing the performance and safety of the battery module **100**.

[0061] In other embodiments, the frame **102** includes the plurality of rows **904**, which may be configured as U-shaped rows or columns, each designed to individually house a row or column of the pack of Li-ion cells **300**, with all the rows or columns integrated into a single, frame structure as the frame **102**. The integration of the plurality of rows **904** multiple U-shaped columns within the frame **102** ensures that each row of Li-ion cells is securely encased within its own U-shaped contour, providing dedicated support and protection. In other embodiments, each of the plurality of rows **904** may include a dedicated compression plate **400**, as well as a dedicated top cover **106**, battery terminals **108**, back cover **600**, relief cap **604**, shims **700**, electric busbars **800**, and combinations thereof. This integration of the plurality of rows **904** with Li-ion cells makes the multi-pack battery module **900** suitable for a wide range of applications, from compact portable devices requiring minimal power to extensive, large-scale energy storage systems necessitating substantial power capacity.

[0062] The frame **102** may be fabricated as a monolithic structure, utilizing manufacturing processes such as casting, sheet metal operations, or additive manufacturing (3D printing). This design approach allows for the frame **102** to be produced from a single material piece, which can simplify the manufacturing process by reducing the number of components and assembly steps required. The design of the frame **102**, whether composed of individual components or as a singular cast or fabricated entity, supports various internal configurations. This includes housing multiple Li-ion cells **200** and integrating components for thermal management, electrical interconnections, and protection mechanisms. The design flexibility accommodates specific requirements for energy storage applications, from portable devices to large-scale energy storage systems.

[0063] Referring to FIGS. **14** & **15**, alternative configurations of the frame **102** are illustrated, according to alternative embodiments of the disclosure. FIG. **14** illustrates a unitary frame **1100** design capable of integrating one of the plurality of rows **904** for a single pack of Li-ion cells **300** and ancillary components into a cohesive structure, suitable for compact energy storage applications. The unitary frame **1100** in FIG. **14** may be the frame **102** of the battery module **100** illustrated in FIGS. **1-2**, **6-7**, and **9-10**. FIG. **15** illustrates a modular frame **1200** comprising the plurality of rows **904**, each designed to house a plurality of the packs of Li-ion cells **300**, offering scalability for modular energy storage solutions. The modular frame **1200** in FIG. **15** may be the frame **102** of the battery module **100** illustrated in FIGS. **11-12**. The front cover **104** and/or the back cover **600** may be formed with the one or both of the ends of the unitary frame **1100** and the modular frame **1200**. The front cover **104** and/or the back cover **600** may also be formed separately and configured to attach to the ends of the unitary frame **1100** and the modular frame **1200**. The frame **102** may be chosen from a U-shaped frame, the unitary frame **1100**, and the modular frame **1200**, whereby the battery module **100** may enclose the frame **102** by using at least one of: the front cover **104**, the back cover **600**, and the top cover **106**, with all other components attached and/or assembled for the battery module **100**.

[0064] In certain embodiments, the modular frame **1200** is engineered to encase a plurality of packs of Li-ion cells **300** along with a plurality of compression plates **400**, where the compression plate **400** may be inserted atop the cell array to apply a uniform compressive force, ensuring consistent electrical contact and structural stability across the cell assembly within the frame **102**. While the compression plate **400** applies uniform pressure across the Li-ion cells **300**, alternative embodiments featuring adjustable or multi-segmented compression plates could offer differential pressure zones. This adaptability would be especially beneficial for modules containing cells of varying capacities, ages, or performance levels.

[0065] From the foregoing, it can be seen that the technology disclosed herein has industrial applicability in a variety of settings such as, but not limited to renewable energy, electric vehicle manufacturing, portable electronic devices, industrial machinery, and grid storage system.

Claims

1. A battery module comprising: a frame; a plurality of lithium-ion cells provided within the frame; a compression plate attached to the frame configured to apply a uniform compression pressure to the plurality of lithium-ion cells in the frame; and a stopper for holding the compression plate at a holding location for maintaining pressure.

2. The battery module of claim 1, wherein the plurality of lithium-ion cells are one of a prismatic cells and pouch cells.

3. The battery module of claim 1, wherein the stopper is sealed to a back cover for maintaining the compression plate in position to maintain a compression pressure.

4. The battery module of claim 1, wherein the frame includes a plurality of cooling channels.

5. The battery module of claim 1, wherein the frame is one chosen from a U-shaped frame, a unitary frame, and a modular frame, and includes at least one of a front cover, a back cover, and a top cover for enclosing the frame.

6. The battery module of claim 1, wherein the front cover includes a plurality of battery terminals.

7. The battery module of claim 1, further comprising a shim provided between the compression plate and the back cover.

8. The battery module of claim 7, wherein the shim is configured to adapt around the stopper between the compression plate and the back cover.

9. A system comprising: a work machine; a battery module for powering the work machine, the battery module having; a frame; a plurality of lithium-ion cells provided within the frame; a compression plate attached to the frame configured to apply a uniform compression pressure to the plurality of lithium-ion cells in the frame; and a stopper for holding the compression plate at a holding location for maintaining pressure.

10. The system of claim 9, wherein the battery module is a multi-pack lithium-ion battery module having: a battery frame; a plurality of packs of lithium-ion cells provided within the battery frame; a plurality of compression plates provided at one end of the frame configured to apply a uniform compression pressure individually to each of packs of lithium-ion cells in the frame; and a plurality of stoppers attached to each of the plurality of compression plates for holding each compression plate at a holding location for maintaining a compression pressure against the plurality of packs of lithium-ion cells.

11. The system of claim 10, wherein each stopper is sealed to the back cover for maintaining each of the compression plates in position to maintain the compression pressure against each pack of Li-ion cells.

12. The system of claim 10, wherein the battery frame includes a plurality of cooling channels.

13. The system of claim 10, the battery frame includes a plurality of battery terminals and the battery frame is one chosen from a U-shaped frame, a unitary frame, and a modular frame and the battery frame further includes at least one of a front cover, a back cover, and a top cover for enclosing the battery frame.

14. The system of claim 10, wherein a plurality of shims are provided between the plurality of compression plates and the back cover.

15. The system of claim 14, wherein the plurality of shims are configured to adapt around the plurality of stoppers between the compression plate and the back cover.

16. A method for assembling a lithium-ion battery module, the method comprising: positioning a plurality of lithium-ion cells within a frame of a battery module; installing a front cover on a first end of the frame, a back cover on a second end of the frame, and a compression plate between the plurality of lithium-ion cells and the back cover; and securing the compression plate in place, via a stopper, to apply a compression pressure on the plurality of lithium-ion cells to mitigate bulging of the plurality of lithium-ion cells within the frame.

17. The method of claim 16, wherein the stopper is connected to the compression plate, further comprising: applying a force, via piston to the stopper to position the compression plate for applying a desired compression pressure on the plurality of lithium-ion cells; and sealing the stopper to the back cover.

18. The method of claim 17, further comprising: providing a shim between the compression plate and the back cover, the shim being configured to adapt to the compression plate and the stopper, for maintaining the compression plate in place to apply a desired compression pressure on the plurality of lithium-ion cells.

19. The method of claim 16, further comprising: forming the frame as one chosen from a U-shaped frame, a unitary frame, and a modular frame, and attaching at least one of a front cover, a back cover, and a top cover for enclosing the frame; forming a plurality of cooling channels in the frame; and installing a plurality of battery terminals in the front cover.

20. The method of claim 16, further comprising: using a work machine powered by the lithium-ion battery module.